A bounded σ-closed operator whose entire resolvent fails σ-continuity

Candidate counterexample packet

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Status: candidate counterexample likely valid; full negative answer to the resolvent question in arXiv:0903.1001. On an explicit norming dual pair, an everywhere-defined bounded σ-closed involution A satisfies $\rho(A) = \mathbb{C} \setminus \{-1, 1\}$ but $\rho_\sigma(A) = \varnothing$.

1 Source question

For a norming dual pair (X, Y), write $\sigma = \sigma(X, Y)$. If A is σ-closed, Kunze defines

$$\rho_\sigma(A) = \{\lambda \in \rho(A) : R(\lambda, A) \text{ is } \sigma\text{-continuous}\}$$

and asks whether $\rho_\sigma(A) = \rho(A)$ must always hold [1, Section 3]. The answer is no, even when A is bounded, everywhere defined, and invertible.

2 Proof Intuition

A bounded operator is $\sigma(X, Y)$-continuous exactly when its adjoint leaves Y invariant. Closedness of its graph is weaker: it is enough that a separating core inside Y be invariant under the adjoint. We exploit this gap on $X = \ell^1$.

Take Y to contain the usual norming subspace c_0 and one extra bounded sequence h. An adjacent-coordinate permutation preserves c_0, so its graph is detected as closed by the coordinate functionals. But it sends h outside Y, destroying weak continuity. Because the permutation is an involution, every resolvent is an affine combination of the identity and the same permutation, and the defect persists throughout the resolvent set.

3 The norming dual pair

Let

$$X = \ell^1(\mathbb{N}), \qquad h = (1, 0, 1, 0, \ldots) \in \ell^\infty, \qquad Y = c_0 + \mathbb{C}h \subset \ell^\infty,$$

with the canonical pairing $\langle x, y\rangle = \sum_{n \geq 1} x_n y_n$ and the inherited norms.

Lemma 1. *The pair (X, Y) is a norming dual pair. Moreover, Y is a closed proper subspace of $X^* = \ell^\infty$.*

Proof. The space c_0 is closed in ℓ^∞, and $h \notin c_0$; hence its one-dimensional extension $Y = c_0 + \mathbb{C}h$ is closed. Since $c_0 \subset Y$, finitely supported sign sequences show that

$$\|x\|_1 = \sup\{|\langle x, y\rangle| : y \in Y,\ \|y\|_\infty \leq 1\}.$$

Conversely, every $y \in Y \subset \ell^\infty = (\ell^1)^*$ has dual norm exactly $\|y\|_\infty$. Thus the two spaces norm each other. Finally, the constant sequence $\mathbf{1}$ does not belong to Y: if $\mathbf{1} = z + ch$ with $z \in c_0$, the odd coordinates force $c = 1$, while the even coordinates would force $z_{2n} = 1$, a contradiction. $\square$

4 Counterexample

Define the adjacent-swap isometry $P : X \to X$ by

$$(Px)_{2n-1} = x_{2n}, \qquad (Px)_{2n} = x_{2n-1} \qquad (n \geq 1).$$

Its Banach adjoint P^* performs the same permutation on ℓ^∞.

Theorem 2. *The bounded operator $A = P$ is $\sigma(X, Y)$-closed, but*

$$\rho(A) = \mathbb{C} \setminus \{-1, 1\}, \qquad \rho_\sigma(A) = \varnothing.$$

In particular, $\rho_\sigma(A) \neq \rho(A)$.

Proof. First observe that $P^* c_0 = c_0$. Suppose a net satisfies $x_\alpha \to x$ and $P x_\alpha \to z$ in $\sigma(X, Y)$. For every $y \in c_0$,

$$\langle z, y\rangle = \lim_\alpha \langle P x_\alpha, y\rangle = \lim_\alpha \langle x_\alpha, P^* y\rangle = \langle x, P^* y\rangle = \langle Px, y\rangle.$$

Because c_0 separates the points of ℓ^1, $z = Px$. Thus the graph of P is $\sigma \times \sigma$ closed.

On the other hand,

$$P^* h = (0, 1, 0, 1, \ldots) = \mathbf{1} - h \notin Y.$$

Indeed, membership of $\mathbf{1} - h$ in $c_0 + \mathbb{C}h$ would imply membership of $\mathbf{1}$. Therefore P is not σ-continuous, by the standard adjoint-invariance criterion for norming dual pairs recalled in the source.

Since $P^2 = I$, its spectrum is $\{-1, 1\}$ and for $\lambda \notin \{-1, 1\}$,

$$R(\lambda, P) = (\lambda I - P)^{-1} = \frac{\lambda I + P}{\lambda^2 - 1}.$$

Applying the adjoint to h gives

$$R(\lambda, P)^* h = \frac{\lambda h + P^* h}{\lambda^2 - 1} = \frac{\mathbf{1} + (\lambda - 1)h}{\lambda^2 - 1}.$$

This sequence is not in Y: all of its even coordinates have the same nonzero constant value $(\lambda^2 - 1)^{-1}$, whereas the even coordinates of every element of $c_0 + \mathbb{C}h$ converge to zero. Hence no ordinary resolvent of P is σ-continuous, proving the claim. □

5 Scope and novelty

The counterexample uses a proper closed norming subspace $Y \subset X^*$. That is essential to this mechanism: for the full dual pair (X, X^*), every bounded operator is weakly continuous. No semigroup, positivity, or unbounded-domain pathology is involved; failure occurs already for a surjective linear isometry of order two.

A bounded search through 11 August 2026 covered the run's result, attempt, and proof-gap indexes and searches for the exact question, the notation $\rho_\sigma(A)$, σ-closed resolvents, and norming dual pairs. It found the original question and subsequent semigroup literature, but no later resolution or matching counterexample. Novelty confidence is moderate pending specialist review; the construction itself is elementary and fully explicit.

References

[1] M. Kunze, *Continuity and equicontinuity of transition semigroups*, Semigroup Forum 79 (2009), 540–560; arXiv:0903.1001. https://arxiv.org/abs/0903.1001.